ClauseGuard

AI-Powered HR Policy Compliance and Workflow Automation Platform

Project Report · Updated Product Version · October 2026

Project Category	Artificial Intelligence, HR Technology, Enterprise SaaS, Document Intelligence
Technology Stack	React, Vite, Python, FastAPI, Supabase PostgreSQL, Qdrant, Ollama, Llama 3.2, LangChain, Sentence Transformers
Status	Working development-stage MVP

> **ClauseGuard turns complex HR policies into actionable, explainable compliance decisions, helping organizations automate routine reviews while keeping humans in control.**

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1. Executive Summary

ClauseGuard is an AI-powered HR policy compliance platform designed to help organizations manage employee requests, interpret internal policies, and streamline compliance-related decisions.

Organizations often rely on lengthy policy documents, manual approval processes, emails, and spreadsheets to manage employee requests. This can result in inconsistent decisions, delayed approvals, difficulty locating relevant policy clauses, and limited visibility into organizational compliance.

ClauseGuard addresses these challenges by combining document intelligence, Retrieval-Augmented Generation (RAG), deterministic rule-based evaluation, and workflow automation in a unified platform.

HR teams can upload and manage company policies, while employees can submit requests and track their status. The system extracts supported rules from policies, evaluates leave requests against those rules, retrieves relevant policy evidence, and generates AI-assisted explanations. HR personnel retain control over decisions requiring human judgment.

The platform also provides role-based access, authentication, policy version management, review workflows, and analytics derived from actual submission records.

ClauseGuard is being developed as a modular enterprise SaaS solution that can evolve from leave-policy compliance into a broader platform for organizational policy governance and workflow automation.

2. One-Line Pitch

> **ClauseGuard turns complex HR policies into actionable, explainable compliance decisions, helping organizations automate routine reviews while keeping humans in control.**

Short elevator pitch

ClauseGuard is an AI-powered HR compliance platform that transforms company policies into searchable knowledge and actionable rules. It helps employees understand policies, automates routine leave-request evaluations, provides evidence-backed explanations, and gives HR teams a centralized system for managing reviews and monitoring compliance.

3. The Problem Statement

3.1 Existing challenges

Many organizations manage employee policies through static PDF documents, shared folders, emails, and manual approval workflows. Although these approaches may work for small teams, they become inefficient as the organization grows.

#	Problem	Description	Impact
1	**Difficult policy interpretation**	HR policies can contain complex conditions, exceptions, eligibility requirements, and approval procedures. Employees may struggle to identify the exact rule relevant to their situation.	Repeated questions, confusion, and unnecessary dependence on HR personnel.
2	**Manual and inconsistent evaluations**	HR teams often review requests individually and compare them with policy documents. Different reviewers may interpret the same rule differently.	Slow processing, inconsistent application of rules, and increased administrative effort.
3	**Lack of explainability**	Employees may receive an approval, rejection, or request for further review without understanding which policy condition influenced the outcome.	Reduced transparency, employee dissatisfaction, and additional clarification requests.
4	**Fragmented workflows**	Employee requests, policy documents, approval decisions, and review records may be distributed across different tools.	Poor visibility, difficult tracking, and inefficient record management.
5	**Limited compliance visibility**	HR managers may lack a consolidated view of request volumes, approval rates, pending reviews, and recurring compliance issues.	Difficulty identifying operational bottlenecks and making informed decisions.

#	Problem	Description	Impact
6	**Policy updates create additional work**	When an organization updates a policy, employees and HR personnel need to identify the latest version and apply its rules consistently.	Risk of outdated interpretations and confusion about which policy version applies.

4. Proposed Solution

ClauseGuard provides a centralized platform that connects policy documents, employee requests, automated evaluations, AI explanations, and HR review workflows.

The system follows a hybrid approach:

1. **Document intelligence:** Extracts text from supported policy documents and divides it into searchable chunks.
2. **Semantic retrieval:** Uses embeddings and vector search to retrieve relevant policy passages.
3. **Rule extraction:** Identifies supported compliance conditions from policy text using deterministic extraction logic.
4. **Compliance evaluation:** Evaluates applicable conditions using a deterministic rule engine.
5. **AI-assisted explanation:** Uses a local language model to explain the evaluation and its supporting policy evidence.
6. **Human review:** Allows HR personnel to review requests and make decisions when intervention is required.
7. **Analytics:** Summarizes real submission records to provide visibility into workflow outcomes.

> **Core principle:** The rule engine determines the compliance outcome, while the language model explains it. The model is not intended to independently override the deterministic evaluation.

5. Product Objectives

The primary objectives of ClauseGuard are to:

- Simplify access to organizational HR policies.
- Reduce repetitive policy-related questions.
- Standardize routine leave-request evaluations.
- Improve transparency through policy citations and explanations.
- Centralize employee submissions and HR review workflows.
- Maintain controlled access through authentication and role-based permissions.
- Support policy versioning and lifecycle management.
- Provide HR analytics using actual application data.
- Establish a foundation for a scalable enterprise SaaS product.

6. Target Users and Customers

Segment	How they use or benefit from ClauseGuard
6.1 HR managers and administrators	Upload policies, manage policy versions, inspect employee requests, review exceptions, make decisions, and monitor workflow outcomes.
6.2 Employees	Ask questions about policies, submit requests, and track their submission status.
6.3 Small and medium-sized businesses	Organizations without extensive HR technology infrastructure can benefit from centralized policy management and structured request workflows.
6.4 Growing enterprises	Larger organizations may use ClauseGuard as a foundation for policy governance, standardized compliance workflows, and cross-team visibility.
6.5 Future customer segments	HR consulting firms, organizations with distributed workforces, and businesses operating under complex internal policy requirements.

7. Existing Features and Current Implementation

The following features reflect the latest implementation state established during development. Individual integrations and end-to-end flows should still be regression-tested before a formal release.

7.1 Authentication and access control

ClauseGuard includes an authentication system based on JWT tokens and password hashing.

Implemented capabilities include:

- User registration and login endpoints.
- Password hashing using bcrypt.
- JWT-based authentication with a 24-hour token lifetime.
- Employee and HR roles.
- Protected frontend routes.
- Backend authorization checks for role-restricted operations.
- Organization ownership checks on relevant HR endpoints.
- Access to organization employee records through authenticated requests.

The frontend uses authentication state to direct HR users to the HR dashboard and employees to the employee dashboard.

Purpose: Restrict access to appropriate application functionality and establish a foundation for multi-user operation.

7.2 HR policy management

HR users can upload supported policy documents, including PDF and TXT files.

Implemented capabilities include:

- Policy document upload.
- Text extraction and indexing.
- Policy metadata storage in PostgreSQL.
- Automatic versioning for policy documents with the same name.
- Archiving of the previous active version when a new version is uploaded.
- Manual activation and archiving of policy versions.

- Policy-specific vector indexing.
- Removal of previous policy chunks before reindexing that policy.

The system distinguishes between active and archived versions, supporting more organized policy lifecycle management.

Purpose: Maintain a centralized source of policy knowledge while making updates manageable.

7.3 Document processing and semantic indexing

ClauseGuard processes supported policy files into searchable text.

The current pipeline uses:

- Document loaders for PDF, TXT, and DOCX text extraction.
- LangChain's RecursiveCharacterTextSplitter.
- A configured chunk size of approximately 1,000 characters and overlap of 150 characters.
- Sentence Transformers using `all-MiniLM-L6-v2`.
- 384-dimensional embeddings.
- Qdrant as the vector database.

Each indexed chunk includes metadata such as policy ID, source, and document type. The metadata allows retrieval to be scoped to the relevant policy instead of searching indiscriminately across all indexed documents.

7.4 AI-powered policy chatbot

The ClauseGuard chatbot allows users to ask natural-language questions about company policies.

Implemented capabilities include:

- Natural-language question submission.
- Semantic retrieval of relevant policy passages.
- Retrieval filtered by policy ID.
- Context-grounded response generation.
- Policy source information and supporting evidence.
- A local Llama 3.2 model accessed through Ollama.
- A FastAPI chat endpoint.

The retrieval pipeline uses similarity filtering to reduce the likelihood of including irrelevant passages. The intended response behavior is to answer from retrieved policy evidence rather than inventing unsupported organizational rules.

Purpose: Help employees and HR personnel understand policies without manually searching through entire documents.

7.5 Automatic policy rule extraction

ClauseGuard includes deterministic rule extraction logic that identifies supported compliance conditions from policy text.

The current extraction implementation supports conditions such as:

- Annual leave limits.
- Maximum or relevant leave durations.
- Advance-notice requirements.
- Emergency-leave conditions.
- Manager-approval requirements.

Extracted rule sets are associated with policy information and saved for use by the evaluation pipeline. This is currently a rule-pattern-based extraction approach rather than a fully general-purpose legal or policy understanding system.

Purpose: Convert supported policy statements into structured conditions that software can evaluate consistently.

7.6 Deterministic compliance engine

The compliance engine evaluates leave requests against structured rules.

Implemented evaluation areas include:

- Leave duration and applicable limits.
- Advance-notice requirements.
- Emergency-leave conditions.
- Annual leave limits.
- Invalid or inconsistent leave dates.
- Conditions requiring further review.

The engine produces structured evaluation results, including statuses, severity information, and explanations of applicable conditions. Requests that cannot be safely or confidently resolved by the available rules can be directed for HR review.

> **Important design principle:** The deterministic engine is the authority for compliance outcomes. Generative AI does not independently change the engine's status or severity.

7.7 AI-assisted review explanations

ClauseGuard combines deterministic evaluation results with language-model-generated explanations. When a leave request is evaluated, the explanation service uses the evaluation result and relevant context to produce a readable summary.

The prompt instructs the language model to:

- Explain the existing decision rather than make a new one.
- Respect the deterministic result.
- Avoid inventing rules or calculations.
- Describe the reason for review when relevant.

This separates the system's decision logic from its explanation layer.

Purpose: Make automated evaluations easier for employees and HR personnel to understand.

7.8 Employee request submission and tracking

The employee-facing application connects to the backend and retrieves actual submission records rather than relying solely on demonstration data.

Implemented workflow capabilities include:

- Employee authentication.
- Submission of leave requests.
- Storage of submission records.
- Display of request status.
- Retrieval of employee submissions from the backend.
- Propagation of review outcomes to the corresponding submission status.

This creates a structured path from request creation to evaluation and review.

7.9 HR review dashboard

The HR dashboard provides a centralized interface for managing employee requests.

Implemented capabilities include:

- Access to review-related records.
- Visibility into employee submissions.
- Display of approval and review statuses.
- HR decision operations.
- Employee record retrieval.
- Organization-scoped employee access.
- Review outcome synchronization with submission records.

The dashboard is designed to help HR personnel identify requests requiring attention and manage them from a common interface.

7.10 Employee directory

The HR dashboard retrieves employee records from the backend. The directory displays real employee information, including:

- Employee name.
- Email address.
- Role.
- Submission-related counts derived from available records.

The earlier placeholder-based employee performance and recognition presentation has been removed from the relevant dashboard view.

Purpose: Provide a reliable employee overview based on stored application data rather than fabricated profile metrics.

7.11 HR analytics dashboard

The latest analytics work replaces several placeholder widgets with information derived from actual submission records.

KPI cards

KPI	Definition
Total Requests	Number of submission records available to the dashboard.
Approval Rate	Approved requests divided by total requests.
Needs Review	Requests currently classified as needing review.
Resolution Rate	Approved and rejected requests divided by total requests.

Weekly submission activity. A seven-day chart visualizes requests submitted over the last seven calendar days, based on submission dates. The chart is titled "Requests Submitted — Last 7 Days."

Compliance outcome breakdown. The dashboard also displays four outcome categories: Approved, Needs Review, Rejected, and Pending. Each category shows its count, proportion of total submissions, and a progress bar.

These analytics are derived from actual submission data returned by the application, with status normalization for supported backend and display labels.

> **Current limitation:** These are operational workflow metrics, not yet a complete compliance-risk intelligence or predictive analytics system.

7.12 Backend API

ClauseGuard uses FastAPI to expose its application functionality. The backend includes routes for authentication, policy management, submissions and reviews, employee records, and chatbot interactions.

Pydantic schemas support request validation and structured responses.

SQLAlchemy is used in parts of the application, while the database utility also provides a Psycopg-based PostgreSQL connection function. These database access patterns should be reviewed for consistency as the application evolves.

The API documentation is available through FastAPI's Swagger interface at `/docs` during local development.

7.13 Existing technology integrations

The current development environment uses:

Layer	Technology
Persistent application data	Supabase PostgreSQL
Semantic vector retrieval	Qdrant
Local language-model inference	Ollama
Embeddings	Sentence Transformers
Frontend	React and Vite
Backend APIs	FastAPI

The existing services have been brought back online and individually verified during the current development session. The backend has also started successfully after explicitly selecting the project's Python 3.11 virtual environment.

8. End-to-End System Workflow

The following workflows describe how the major components work together.

8.1 Policy onboarding

1. An authorized HR user uploads a supported policy document.
2. The backend extracts the document's text.
3. The policy is stored with its metadata and version information.
4. The rule extractor identifies supported compliance conditions.
5. The policy text is divided into overlapping chunks.
6. The embedding model converts those chunks into vectors.
7. The vectors and their metadata are stored in Qdrant.
8. The policy becomes available to the relevant retrieval and evaluation workflows.

8.2 Employee leave request

1. The employee signs in.
2. The employee submits a leave request.

3. The backend retrieves the applicable policy rules.
4. The deterministic engine evaluates the request.
5. Relevant policy evidence is retrieved where applicable.
6. The explanation service produces a readable explanation of the result.
7. The request and evaluation are recorded.
8. The employee can view the resulting status.
9. HR can inspect requests requiring review and make appropriate decisions.

8.3 Policy question and answer

1. A user submits a natural-language policy question.
2. The system generates a query embedding.
3. Qdrant retrieves relevant policy passages.
4. Retrieval filtering selects suitable evidence.
5. The language model receives the question and retrieved context.
6. The system generates a context-grounded answer.
7. The response includes source information and supporting evidence where available.

8.4 HR analytics

1. The dashboard retrieves available submission records.
2. It normalizes supported status representations.
3. It calculates request totals, approval rate, review counts, and resolution rate.
4. It aggregates submissions into seven calendar-day buckets.
5. It presents the compliance outcome distribution.

This design makes the analytics responsive to the application's data instead of relying on fixed demonstration values.

9. Technical Architecture

9.1 Architecture overview

Layer	Technology	Responsibility
Presentation	React and Vite	Employee and HR interfaces, authentication screens, policy-related interactions, request tracking, review management, and analytics.
Application	FastAPI	API requests, authentication, authorization, policy operations, compliance evaluations, and integration between the frontend and AI components.
Relational data	Supabase PostgreSQL	Structured application records such as users, organizations, policy metadata, submissions, reviews, and supporting relationships.
Vector retrieval	Qdrant	Embedded policy chunks with metadata for semantic retrieval.
Embedding	Sentence Transformers	Transforms policy passages and user questions into 384-dimensional vectors using `all-MiniLM-L6-v2`.
Language model	Ollama and Llama 3.2	Generates policy-grounded answers and readable explanations from supplied context.
AI orchestration	LangChain and custom Python modules	Coordinates document chunking, retrieval, and language-model interaction.

9.2 Architecture diagram

```
Browser-based React frontend
        ↓
FastAPI backend and authorization
        ↓
Application services and deterministic rule engine
        ↓
Supabase PostgreSQL for structured records
        ↓
Qdrant for policy-vector retrieval
        ↓
Sentence Transformers for embeddings
        ↓
Ollama / Llama 3.2 for grounded explanations
```

The arrows represent logical relationships rather than a strict sequential path. For example, a leave request can be evaluated by the rule engine without requiring a language-model call for the decision itself.

9.3 Why this architecture?

The architecture separates structured data, semantic search, business rules, and generative explanations. This provides several advantages:

- Relational storage for consistent application records.
- Vector search for meaning-based document retrieval.
- Deterministic logic for repeatable rule evaluation.
- Local inference for greater control over model execution.
- Modular components that can be tested and upgraded independently.

10. Database and Information Management

Supabase PostgreSQL is the persistent relational database for the application. The database layer supports organizational and user records, policy metadata, submissions, review outcomes, and associated information.

Qdrant stores policy chunks and their metadata separately from the relational database. This separation avoids using a vector database as a substitute for transactional application storage.

Planned database improvements

The following remain engineering priorities:

- Formalize and document all schema relationships.
- Introduce version-controlled database migrations using Alembic.
- Review transaction boundaries between request evaluation, review creation, and status updates.
- Standardize database access patterns.
- Strengthen organization isolation across all relevant routes and data queries.
- Establish a reliable audit-history model.
- Define retention and deletion policies for uploaded documents and derived embeddings.

11. Security, Privacy, and Responsible AI

ClauseGuard is intended to handle internal organizational documents and employee requests. Security must therefore be part of the core design.

11.1 Existing security measures

The current implementation includes:

- Password hashing with bcrypt.
- JWT-based authentication.
- Role-based authorization for protected operations.
- Organization ownership checks on relevant HR routes.
- Protected frontend routes.
- Policy-specific retrieval filtering.
- A deterministic compliance engine separated from AI-generated explanations.

11.2 Security work still required

Before a production release, the following should be completed and verified:

- Move hardcoded development secrets into environment configuration and rotate them where necessary.
- Validate authentication and authorization across every sensitive endpoint.

- Ensure production database credentials are not exposed in client-side code or version control.
- Configure HTTPS and appropriate deployment security settings.
- Review file upload validation, file size limits, and malicious-file handling.
- Enforce organization-level data isolation consistently.
- Evaluate Supabase Row Level Security and other appropriate database protections.
- Add rate limiting and abuse protection.
- Implement secure session expiration and logout behavior.
- Define employee-data retention, access, and deletion policies.
- Test that unauthorized users cannot access another organization's documents or submissions.

11.3 Responsible AI principles

ClauseGuard follows a hybrid AI approach:

- The rule engine makes the compliance evaluation.
- Retrieval supplies relevant policy context.
- The language model explains rather than overrides the result.
- Unsupported questions should be handled cautiously.
- Ambiguous cases should be escalated to a human reviewer.
- Citations and evidence should be traceable to source documents.

The goal is not to replace HR judgment, but to make routine processing more consistent, transparent, and efficient.

12. Current Limitations

The present version is a working development-stage product, not a fully production-ready enterprise platform.

Key limitations include:

1. Rule extraction supports defined policy patterns and is not guaranteed to interpret every policy formulation.
2. Compliance evaluation currently focuses on leave-related rules rather than every possible HR workflow.

3. RAG quality depends on document extraction, retrieval relevance, and the quality of the underlying policy.
4. AI-generated explanations require validation against the actual evaluation and evidence.
5. Authentication and access control need a complete security audit.
6. Advanced predictive analytics and proactive risk detection have not yet been implemented.
7. Production-grade deployment, monitoring, automated backups, and disaster recovery require additional work.
8. Formal audit history and comprehensive multi-organization isolation need further validation.
9. Performance and scalability need to be tested using realistic organizational workloads.

These limitations define the next phase of development rather than undermining the existing foundation.

13. Planned Features and Product Roadmap

The following items are proposed enhancements, not claims about functionality already delivered.

Phase	Focus	Priority
1	Stabilization and reliability	Immediate
2	Compliance Risk Overview	High
3	Improved policy intelligence	High
4	Expanded HR workflows	High
5	Auditability and organizational governance	High
6	Enterprise SaaS capabilities	Medium to high
7	Advanced analytics and intelligent assistance	Future

Phase 1 — Stabilization and reliability

Priority: Immediate

- Test login and registration end to end.
- Validate employee request submission and status tracking.
- Test policy upload, versioning, activation, and archiving.
- Validate RAG answers and source citations.
- Test deterministic rules against edge cases.
- Verify HR review decisions propagate correctly.
- Remove development secrets and debug artifacts.
- Review database migrations and schema consistency.
- Add automated integration and regression tests.

Expected outcome: A reliable and demonstrable end-to-end MVP.

Phase 2 — Compliance Risk Overview

Priority: High

Develop an HR dashboard section that summarizes operational compliance risk using explainable indicators.

Potential metrics include:

- Requests awaiting HR review.
- Requests approaching defined deadlines.
- Frequently triggered policy rules.
- Policies with high request-related exception rates.
- Departments with unusual request patterns, where reliable department data is available.
- Changes in review volume over time.

The initial version should use transparent thresholds and real application records. It should not present a heuristic score as a validated prediction.

Expected outcome: Better visibility into operational bottlenecks and review priorities.

Phase 3 — Improved policy intelligence

Priority: High

- Expand deterministic rule extraction coverage.
- Support more policy formats and structured rule types.

- Add better handling of exceptions and conditional clauses.
- Improve retrieval ranking and evidence selection.
- Add confidence-aware fallback behavior.
- Provide clearer citations that identify the exact supporting passage.
- Validate extracted rules with HR approval before activation.

Expected outcome: More reliable interpretation and easier verification of policy-derived rules.

Phase 4 — Expanded HR workflows

Priority: High

- Extend beyond leave requests to selected expense and reimbursement workflows.
- Introduce configurable approval chains.
- Add request assignment and reviewer comments.
- Add notifications for pending decisions.
- Support escalation when requests remain unresolved.
- Provide more detailed decision histories.

Each new workflow should have its own explicit rules, validations, and test cases.

Expected outcome: A broader HR workflow automation platform.

Phase 5 — Auditability and organizational governance

Priority: High

- Add an immutable or tamper-evident audit history where appropriate.
- Record policy versions used for evaluations.
- Record rule versions, evaluation outcomes, and reviewer decisions.
- Add timestamps and actor information to significant actions.
- Strengthen tenant isolation.
- Introduce organization-specific policy administration.
- Support configurable retention and access controls.

Expected outcome: Improved traceability and suitability for enterprise governance.

Phase 6 — Enterprise SaaS capabilities

Priority: Medium to high

- Organization onboarding and administration.
- Invitations and employee provisioning.
- Role and permission management.
- Subscription plans and seat management.
- Usage and quota monitoring.
- Administrative reporting.
- Production deployment and monitoring.
- Backup and recovery procedures.

Expected outcome: A commercially deployable multi-organization service.

Phase 7 — Advanced analytics and intelligent assistance

Priority: Future

- Department-wise and time-based analytics.
- Policy-change impact analysis.
- Identification of recurring compliance issues.
- Forecasting of HR review workload.
- Improved semantic search and hybrid retrieval.
- Carefully evaluated recommendations for HR prioritization.
- Natural-language questions over authorized organizational metrics.

Any predictive feature should be validated before being used to influence real employment decisions.

14. Innovation and Differentiation

ClauseGuard's principal differentiator is its combination of deterministic compliance evaluation and context-grounded generative AI.

A conventional chatbot may answer questions about a policy but may not execute a structured workflow. A conventional approval system may record a decision but may not explain the relevant policy evidence. ClauseGuard aims to bridge that gap.

Key differentiators

1. **Rules plus RAG.** The platform combines structured evaluation with semantic retrieval instead of relying on a language model alone.
2. **Explainable outcomes.** Evaluations can be paired with policy evidence and human-readable explanations.
3. **Policy lifecycle management.** Policy versions and their associated vector indexes are managed as part of the document workflow.
4. **Human-in-the-loop review.** Ambiguous or exceptional cases can be directed to HR personnel.
5. **Local language-model inference.** Ollama provides a local inference option during development, reducing reliance on an external hosted language-model API. This does not by itself guarantee complete data privacy or production security.
6. **Real operational analytics.** The dashboard is moving away from placeholder widgets toward metrics calculated from actual application records.

15. Business Model

ClauseGuard can follow a subscription-based enterprise SaaS model.

Plan	Intended for
Starter	Small businesses with basic policy management, employee requests, and limited usage.
Professional	Growing organizations that need expanded policy intelligence, HR analytics, and more workflow capacity.
Enterprise	Organizations requiring advanced access controls, configurable workflows, auditability, integrations, and dedicated support.

Possible pricing units include organization size, active employee seats, document-processing volume, and workflow usage.

> These are proposed commercial options. Actual pricing, willingness to pay, operating costs, and market demand have not yet been validated.

16. Expected Benefits and Success Metrics

ClauseGuard aims to reduce administrative effort while improving consistency and transparency.

Potential success metrics include:

- Average time required to process a request.
- Percentage of requests processed without manual clarification.
- Percentage of evaluations with valid supporting evidence.
- HR review backlog and average resolution time.
- Employee understanding and satisfaction.
- Accuracy of extracted policy rules.
- Rate of incorrect or unsupported chatbot responses.
- Percentage of requests correctly escalated for human review.

Targets should be established after collecting a baseline from real or carefully designed pilot workflows. Benefits such as time savings should be measured rather than assumed.

17. Implementation and Evaluation Plan

A structured evaluation will help establish whether ClauseGuard works reliably.

Area	What to test
Functional testing	Registration, login, policy upload, versioning, chatbot interactions, submissions, evaluation, review decisions, and analytics.
Rule-engine testing	Normal requests, notice-period boundaries, annual limits, emergency leave, invalid dates, and ambiguous conditions.
RAG evaluation	Questions with known answers; retrieval relevance, answer correctness, citation validity, and unsupported-answer behavior.
Security testing	Unauthorized access, incorrect roles, cross-organization access attempts, token expiration, upload validation, and sensitive-data exposure.

Area	What to test
Performance testing	Response times, document indexing time, vector retrieval latency, and behavior under multiple concurrent requests.
User acceptance testing	Representative employees and HR reviewers complete realistic workflows and provide feedback on clarity and usability.

18. Pitch Deck — 12-Slide Presentation

Slide 1 — Title

ClauseGuard

From complex HR policies to explainable compliance decisions.

AI-powered policy intelligence and workflow automation.

Slide 2 — The Problem

HR teams face fragmented documents, manual request evaluations, inconsistent policy interpretation, and limited visibility into compliance workflows.

Employees struggle to understand policy requirements, while HR personnel spend time answering repetitive questions and reviewing routine requests.

Key message: Policies are documented, but applying them consistently remains difficult.

Slide 3 — Our Solution

ClauseGuard connects policy documents, AI-powered search, deterministic rule evaluation, and HR workflows in one platform.

- Search and understand company policies.
- Evaluate supported leave-request conditions.
- Explain outcomes with policy evidence.
- Escalate cases requiring human judgment.
- Monitor request outcomes through HR analytics.

Slide 4 — Product Demonstration

Show the actual working application:

1. HR signs in and uploads a policy.
2. The system processes and indexes the document.
3. An employee asks a policy question.
4. An employee submits a leave request.
5. ClauseGuard evaluates the request and explains the outcome.
6. HR reviews the request if necessary.
7. The analytics dashboard updates from actual records.

Use real application screenshots and a tested demonstration flow.

Slide 5 — How It Works

Upload → Extract → Index → Retrieve → Evaluate → Explain → Review

- Document processing prepares policy text.
- Sentence Transformers creates embeddings.
- Qdrant retrieves relevant evidence.
- The deterministic engine evaluates supported conditions.
- Llama 3.2 generates explanations.
- HR handles exceptional or uncertain cases.

Emphasize that the language model explains the decision rather than independently determining compliance.

Slide 6 — Technology

- Frontend: React and Vite
- Backend: Python and FastAPI
- Relational database: Supabase PostgreSQL
- Vector database: Qdrant
- Embeddings: Sentence Transformers
- LLM: Llama 3.2 through Ollama
- Orchestration: LangChain

Highlight the separation between structured business logic, semantic retrieval, and language generation.

Slide 7 — Core Features

- Authentication and role-based access.
- HR policy upload and version management.
- Policy chatbot with retrieved evidence.
- Deterministic leave-compliance evaluation.
- AI-assisted explanations.
- Employee submission tracking.
- HR review and decision workflows.
- Real-data HR analytics.

Slide 8 — Differentiation

ClauseGuard combines capabilities that are often implemented separately:

- Policy understanding through semantic search.
- Repeatable compliance checks through deterministic rules.
- Evidence-backed explanations through RAG.
- Human review for exceptions.
- Centralized workflow tracking.

Positioning: More than a chatbot and more than a conventional approval dashboard.

Slide 9 — Target Market and Business Model

Initial target customers:

- Small and medium-sized businesses.
- Growing organizations with centralized HR operations.
- Enterprises with substantial internal policy documentation.

Proposed revenue model:

- Subscription plans.
- Employee-seat pricing.
- Tiered workflow and document-processing limits.

- Enterprise features and integrations.

Market demand and pricing will need validation through customer interviews and pilot deployments.

Slide 10 — Roadmap

- **Now:** Stabilize and validate the current MVP.
- **Next:** Compliance Risk Overview, better policy extraction, and stronger auditability.
- **Later:** Expense workflows, configurable approvals, notifications, organization administration, and subscription management.
- **Long term:** Enterprise integrations and validated predictive HR operations analytics.

Slide 11 — Impact and Validation

Demonstrate measurable outcomes rather than unverified claims.

Track:

- Request processing time.
- Retrieval and citation correctness.
- Rule-extraction accuracy.
- HR review workload.
- Resolution time.
- Employee satisfaction.

Use pilot results to establish the actual operational impact.

Slide 12 — Closing

ClauseGuard makes organizational policies actionable, explainable, and easier to manage.

We are building a platform where employees receive clearer guidance, HR teams gain structured workflows, and compliance decisions remain transparent and reviewable.

> *Understand the policy. Apply the rule. Explain the decision.*

19. Conclusion

ClauseGuard establishes a foundation for AI-assisted HR policy compliance by combining policy management, semantic retrieval, deterministic rule evaluation, generative explanations, and structured employee workflows.

The current implementation includes the major building blocks of a functional MVP: authentication, policy lifecycle management, RAG-based policy questions, rule-based leave evaluation, HR review workflows, and analytics derived from application data.

The next stage is to strengthen reliability, validate end-to-end behavior, improve security and auditability, and extend the platform into additional HR workflows.

The long-term vision is to make ClauseGuard a trusted policy intelligence and workflow automation platform that helps organizations apply their internal policies more consistently while preserving human oversight for important decisions.

Project status: Working development-stage MVP with additional reliability, security, validation, and productization work remaining.

Core principle: Automate routine compliance checks, explain outcomes with evidence, and keep human judgment where it matters.